

Countable Additivity is Not First-Order Axiomatizable

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Abstract

In the first-order language of Boolean algebras equipped with a normalized finitely additive charge, no first-order theory characterizes those models whose charge extends to a σ -additive measure on the generated σ -algebra. Equivalently, countable additivity is not first-order axiomatizable among finitely additive probability Boolean algebras. The proof is a single ultraproduct argument: Dirac charges on the finite-cofinite algebra are individually σ -additive, but their ultraproduct is the finite-cofinite charge, which fails σ -additivity — contradicting Łoś’s theorem if any first-order axiomatization existed.

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A property of structures is first-order axiomatizable in a language L if there exists a set of L -sentences whose models are exactly the structures satisfying it. Łoś’s theorem gives the key diagnostic: if a property fails in an ultraproduct of structures that individually satisfy it, no first-order axiomatization exists.

Countable additivity distinguishes σ -additive measures from finitely additive charges. That it cannot be expressed by a single first-order sentence is clear: the condition quantifies over countable sequences, unavailable in first-order logic. Its non-first-order character is treated as background knowledge in the probability logic literature [1, 2, 3], but the sharper question — whether countable additivity admits a first-order *axiomatization* by any set of sentences, not merely a single formula — does not appear to have been settled by an explicit proof.

We settle it negatively.

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Let $\mathcal{L}_{\text{BA},\mu}$ be the one-sorted first-order language with:

- Boolean operations $\wedge, \vee, \neg, 0, 1$;
- for each rational $q \in [0, 1] \cap \mathbb{Q}$, unary relation symbols $M_{\leq q}(x)$ and $M_{\geq q}(x)$, intended to mean $\mu(x) \leq q$ and $\mu(x) \geq q$ respectively.

A structure for $\mathcal{L}_{\text{BA},\mu}$ is a Boolean algebra B together with a $[0, 1]$ -valued finitely additive probability μ , encoded through the truth of the predicates $M_{\leq q}$ and $M_{\geq q}$. Using rational comparison predicates rather than a function symbol into $[0, 1]$ avoids introducing a second numeric sort.

Definition 1 (First-order theory of finitely additive probabilities). Let T_{fa} be the $\mathcal{L}_{\text{BA},\mu}$ -theory axiomatizing:

- (i) B is a Boolean algebra;
- (ii) normalization: $\mu(0) = 0$ and $\mu(1) = 1$;
- (iii) monotonicity: $x \leq y \Rightarrow \mu(x) \leq \mu(y)$;
- (iv) finite additivity: for all rationals r, s with $r + s \leq 1$,

$$x \wedge y = 0 \wedge M_{\leq r}(x) \wedge M_{\leq s}(y) \Rightarrow M_{\leq r+s}(x \vee y),$$

and similarly for lower bounds.

The class of Boolean algebras with normalized finitely additive probability is first-order axiomatizable in $\mathcal{L}_{\text{BA},\mu}$. Countable additivity, by contrast, requires quantification over an arbitrary countable sequence $(a_n)_{n \in \mathbb{N}}$ — not available in first-order logic — and so is not a sentence of $\mathcal{L}_{\text{BA},\mu}$.

Proposition 2 (σ -additive extension is not first-order axiomatizable). *There is no first-order $\mathcal{L}_{\text{BA},\mu}$ -theory $T \supseteq T_{\text{fa}}$ such that, for every $(B, \mu) \models T_{\text{fa}}$,*

$$(B, \mu) \models T \iff \mu \text{ extends to a } \sigma\text{-additive measure on } \sigma(B).$$

Equivalently, countable additivity is not first-order axiomatizable among finitely additive probability Boolean algebras.

Proof. Let E be the Boolean algebra of finite-cofinite subsets of $\mathbb{Q}$, fix an enumeration $q : \mathbb{N} \rightarrow \mathbb{Q}$, and let δ_n be the Dirac charge at $q(n)$. Each (E, δ_n) is a model of T_{fa} in which δ_n is σ -additive. Suppose for contradiction that a theory T as described exists; then each $(E, \delta_n) \models T$.

Let $\mathcal{U}$ be a nonprincipal ultrafilter on $\mathbb{N}$ extending the cofinite filter. By Łoś's theorem [4], the ultraproduct $\prod_{\mathcal{U}}(E, \delta_n)$ also models T . The charge induced on the diagonal copy of E is

$$\ell(A) = \lim_{\mathcal{U}} \delta_n(A) = \lim_{\mathcal{U}} \mathbf{1}_{q(n) \in A}.$$

If A is finite, $q(n) \in A$ for only finitely many n , so $\ell(A) = 0$. If A is cofinite, $q(n) \in A$ for cofinitely many n , so $\ell(A) = 1$ since $\mathcal{U}$ extends the cofinite filter. Thus ℓ is the finite-cofinite charge.

The sequence $A_k = \mathbb{Q} \setminus \{q(0), \dots, q(k)\}$ satisfies $A_0 \supseteq A_1 \supseteq \dots$ and $\bigcap_k A_k = \emptyset$, yet $\ell(A_k) = 1$ for all k . Hence the charge on the diagonal copy of E fails σ -additivity and does not extend to a σ -additive measure on $\sigma(E)$. Consequently the ultraproduct cannot satisfy any theory T characterizing σ -additive extension, contradicting Łoś's theorem applied to the assumption that each $(E, \delta_n) \models T$. $\square$

Remark 3. The ultraproduct $\prod_{\mathcal{U}}(E, \delta_n)$ is an $\mathcal{L}_{\text{BA}, \mu}$ -structure in its own right; ℓ is the charge induced on the diagonal copy of E inside it. Identifying the ultraproduct with (E, ℓ) is harmless for the contradiction, since the failure of σ -additivity is witnessed on the diagonal copy.

The proof produces one purely finitely additive charge as an ultraproduct of σ -additive ones; that single witness suffices for the non-axiomatizability conclusion. A stronger question is:

Which purely finitely additive charges on a Boolean algebra arise as pointwise ultralimits of σ -additive probabilities on the same algebra?

On a σ -algebra the answer is none: any pointwise ultralimit of σ -additive probabilities is itself σ -additive, by Nikodym-type convergence [5], IV.9.8. On a general Boolean algebra the situation is more structured, and the problem is settled in several natural regimes. The remaining open case reduces to whether there exists a non- σ -complete non-atomic Boolean algebra admitting no σ -additive probability.

References

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